

Executive Summary & Technical Report: Days 1 & 2

1. Business Context & Problem Formulation

- **Objective:** Predict individual personal loan default (*target* = 1 vs *target* = 0) at the exact time of credit underwriting / loan application (\$t = 0\$).
 - **Data Scale:** 96,112 historical loan applications in training data (54 columns) and 23,888 unlabelled loans in the scoring/test set (53 columns).
 - **Target Distribution & Imbalance:** Baseline default rate of 14.36% (~14.4%), presenting moderate class imbalance requiring stratified evaluation.
 - **Evaluation Metric:** Out-of-fold (OOF) Receiver Operating Characteristic Area Under the Curve (ROC-AUC).
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2. Day 1: Data Audit, Diagnostics & Leakage Discovery

Day 1 focused on data profiling, missingness patterns, and verifying temporal validity (\$t = 0\$).

2.1 Missingness Profiling

- Severe structural missingness observed in delinquency and inquiry lookbacks:
 - *m_since_delinq*: 46,462 missing (~48.3%)
 - *m_since_inquiry*: 9,855 missing (~10.3%)
- Moderate sparsity observed in applicant demographics (*emp_years*: 5,982 missing) and revolving credit card utilization fields (*card_util*: 1,036, *pct_cards_hi_util*: 1,033).

2.2 Critical Finding — Post-Origination Data Leakage

- Auditing the feature space against domain credit logic identified 9 servicing / post-origination fields:
 - *principal_recv*, *payments_total*, *last_txn_amt*, *late_fees*, *residual_amt*, *fee_adj*, *score_recent*, *review_gap_m*, and *account_flag*.
- These features capture borrower repayment behavior, recoveries, and account review events occurring after loan origination (\$t > 0\$).

2.3 Naive Benchmark (The Leakage Trap)

- Fitting an initial baseline model with all raw features produced an artificially inflated validation score of **Mean CV AUC = 0.99974 ± 0.00010** (Fold AUCs: 0.99957, 0.99982, 0.99968, 0.99983, 0.99982).

- *Diagnosis:* The model memorized post-origination servicing history rather than assessing baseline application credit risk. In production, this would cause catastrophic model failure.
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3. Day 2: Tabular Modeling, Clean Validation & Submission Pipeline

Day 2 enforced strict temporal data boundaries, implemented leakage-resistant cross-validation, and produced production-grade predictions.

3.1 Pipeline Sanitation

- Permanently dropped all 9 post-origination servicing features from both training and test matrices prior to model fitting.
- Retained all 43 genuine application-time features (applicant demographics, loan terms, credit bureau inquiries, and tradeline balances).
- Employed LightGBM Classifier, which naturally handles missing numerical values without distorting feature distributions via naive global imputations.

3.2 Validation Strategy

- Implemented **Stratified 5-Fold Cross-Validation** (*random_state=42*) to preserve the 14.4% class ratio across every fold and eliminate single-split optimism.

3.3 Clean Model Performance

- **Fold 1 AUC:** 0.69411
- **Fold 2 AUC:** 0.69493
- **Fold 3 AUC:** 0.69346
- **Fold 4 AUC:** 0.69670
- **Fold 5 AUC:** 0.69896
- **Clean Mean CV AUC:** **0.69563 ± 0.00199**
- The tight standard deviation (± 0.00199) demonstrates high stability and genuine out-of-sample generalization.

3.4 Test Scoring & Serialization

- Out-of-fold probability estimates from the 5 fold models were averaged on *d2assignment_test.csv*.
 - Generated *submission.csv* containing all 23,888 test *row_id* entries with their respective *predicted_probability*, fully adhering to submission guidelines.
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4. Operational Risk & Model Limitations

- **Bureau Feature Dependence:** Model decisions rely heavily on bureau history (*bureau_score*, *open_tradelines*, inquiries). Shifts in macroeconomic conditions or bureau reporting conventions will impact calibration.

Underwriting Policy & Drift: Because post-origination behavioral signals were excluded to guarantee validity at origination, the pipeline requires ongoing drift monitoring (Population Stability Index / KS tests) to detect borrower population shifts over tenure.